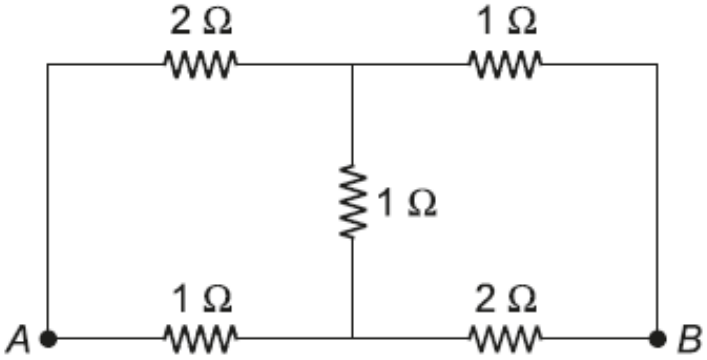
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Find the equivalent resistance between points A and B in the circuit shown in Fig.



(A) 1.4 ohm

(B) 1.8 ohm

(C) 1.6 ohm

(D) 2.4 ohm

Q.2

A conductor of resistance 3Ω is stretched uniformly till its length is doubled. The wire is now bent in the form of an equilateral triangle. The effective resistance between the ends of any side of the triangle in ohms is:

(A) $\frac{9}{2}$

(B) $\frac{8}{3}$

(C) 2

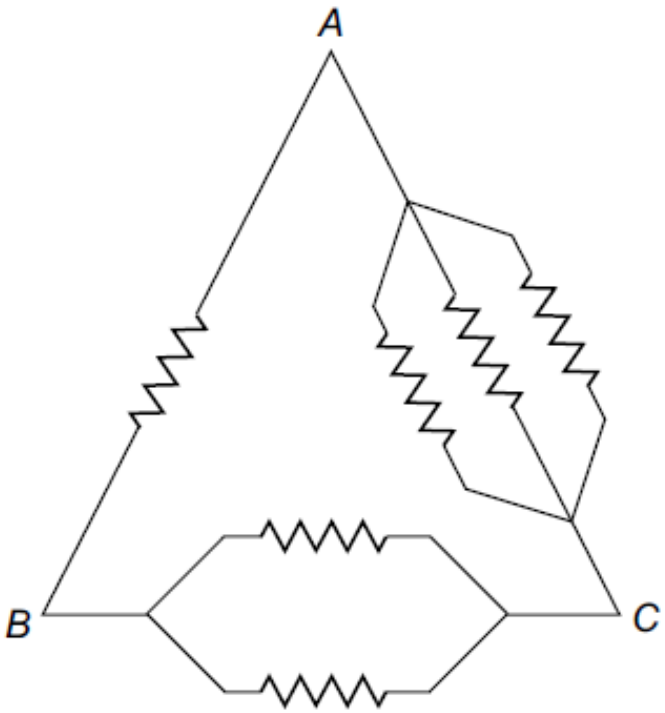
(D) 1

Q.3

Six equal resistances are connected between points A, B and C as shown Fig. If R_1, R_2 and R_3 are the net resistances between A and B, between B and C and between A and C respectively, then $R_1 : R_2 : R_3$

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will be equal to



(A) 6 : 3 : 2

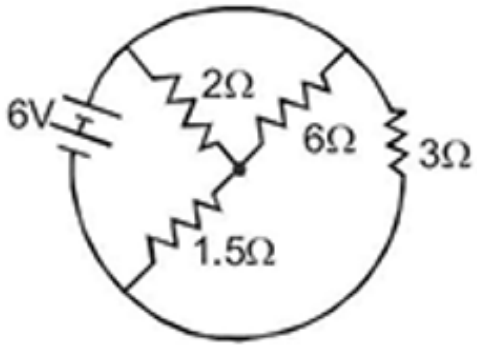
(B) 1 : 2 : 3

(C) 5 : 4 : 3

(D) 4 : 3 : 2

Q.4

The total current supplied to the circuit by the battery is



(A) 4 A

(B) 2 A

(C) 1 A

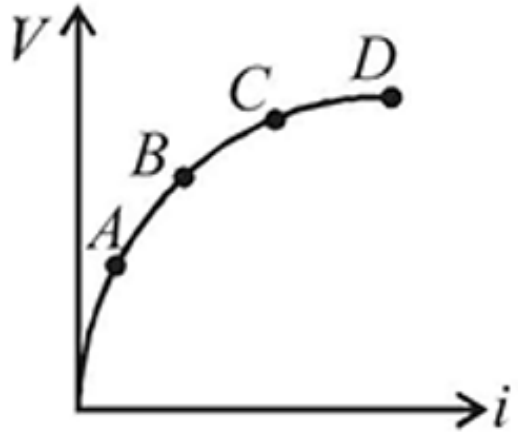
(D) 6 A

Q.5

Variation of current passing through a conductor as the voltage applied across its ends is varied as shown in the

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adjoining diagram. If the resistance (R) is determined at the points A, B, C and D, we will find that



(A) $R_C = R_D$

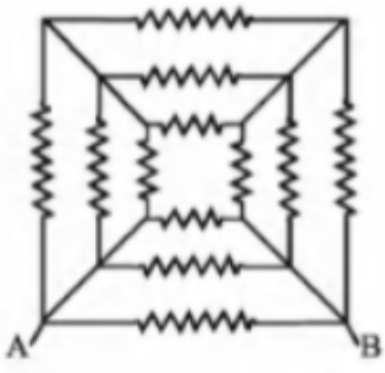
(B) $R_B > R_A$

(C) $R_C > R_B$

(D) $R_A > R_B$

Q.6

Twelve resistors each of resistance 16Ω are connected in the circuit as shown. The net resistance between AB is



(A) 1Ω

(B) 2Ω

(C) 3Ω

(D) 4Ω

Q.7

n equal resistors are first connected in series and then connected in parallel. What is the ratio of the maximum to the minimum resistance?

(A) n

(B) $1/\mathrm{n}^2$

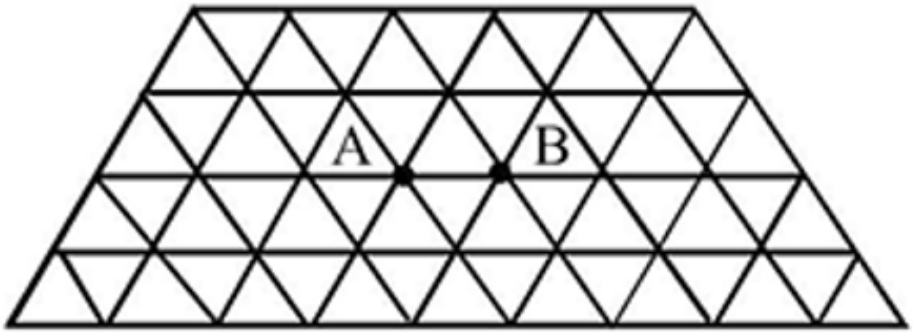
(C) n^2

(D) $1/\mathrm{n}$

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Q.8

There is an infinite wire grid with cells in the form of equilateral triangles. The resistance of each wire between neighbouring joint connections is R_0. The net resistance of the whole grid between the points A and B as shown is



(A) R_0

(B) $\frac{R_0}{2}$

(C) $\frac{R_0}{3}$

(D) $\frac{R_0}{4}$

Q.9

The electric resistance of a certain wire of iron is R. If its length and radius are both doubled, then

(A) the resistance and the specific resistance, will both remain unchanged

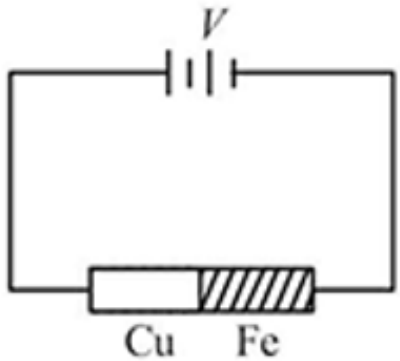
(B) the resistance will be doubled and the specific resistance will be halved

(C) the resistance will be halved and the specific resistance will remain unchanged

(D) the resistance will be halved and the specific resistance will be doubled

Q.10

Two rods are joined end to end, as shown. Both have a cross-sectional area of $0.01\ cm^2$. Each is 1 meter long. One rod is of copper with a resistivity of 1.7×10^{-6} ohm-centimeter, the other is of iron with a resistivity of 10^{-5} ohm-centimeter.
How much voltage (in volt) is required to produce a current of 1 ampere in the rods?



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(A) 0.111

(B) 0.113

(C) 0.115

(D) 0.117

Q.11

Two wires of the same metal have same length, but their cross-sections are in the ratio $3:1$. They are joined in series. The resistance of thicker wire is 10Ω. The total resistance of the combination will be

(A) 10Ω

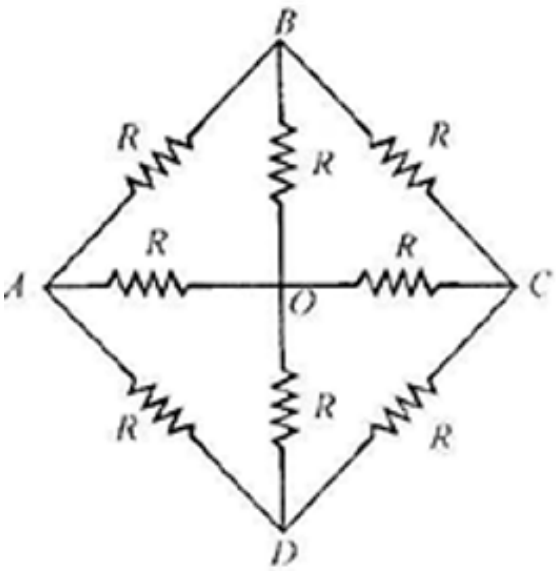
(B) 20Ω

(C) 40Ω

(D) 100Ω

Q.12

In the network shown, each resistance is equal to R. The equivalent resistance between adjacent corners A and D is



(A) R

(B) $\frac{2}{3}\mathrm{R}$

(C) $\frac{3}{7}\mathrm{R}$

(D) $\frac{8}{15}\mathrm{R}$

Q.13

The resistance of the four arms $\mathrm{P}, \mathrm{Q}, \mathrm{R}$ and S in a Wheatstone's bridge are $10\mathrm{ohm}, 30\mathrm{ohm}, 30\mathrm{ohm}$ and 90 ohm, respectively. The e.m.f. and internal resistance of the cell are 7 volt and 5 ohm respectively. If the galvanometer resistance is 50 ohm, the current drawn from the cell will be

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(A) 0.2 A

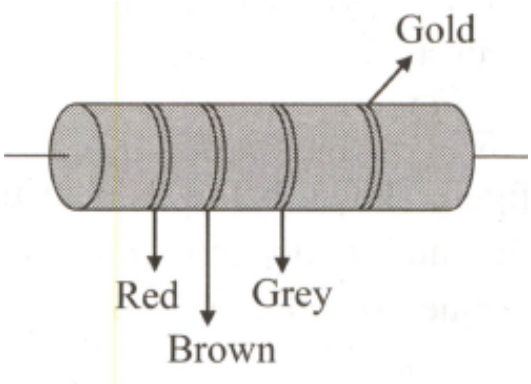
(B) 0.1 A

(C) 2.0 A

(D) 1.0 A

Q.14

The value of the resistance of a carbon resistor has bands of different colours as mentioned in the figure is _____ $G\Omega \pm 5\%$



(A) 2.0

(B) 2.1

(C) 2.2

(D) 2.3

Q.15

A moving coil galvanometer, having a resistance G. produces full scales deflection when a current Ig flows through it. It can be converted into (i) an ammeter of range 0 to Io $(Io > \ Ig)$ by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to $VV = GI_0$by connecting a series resistance Rv to it. Then,choose the correct option

(A) $R_A R_V = G^2\left(\frac{I_0 - I_g}{I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

(B) $R_A R_V = G^2$ and πan $\frac{R_A}{R_v} = \frac{I_g}{(I_0 - I_g)}$

(C) $R_A R_v = G^2$and $\frac{R_A}{R_v} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

(D) $R_A R_V = G^2\left(\frac{I_g}{I_0 - I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g}\right)^2$

Q.16

Two resistances are connected in the two gaps of a meter bridge. The balance point is 20 cm from the zero

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end. When a resistance 15 Ω is connected in series with the smaller of two resistance, the null point get shifted to 40 cm. What is the smaller of the two resistance values ?

(A) 8 Ω

(B) 9 Ω

(C) 10 Ω

(D) 12 Ω

Q.17

A potentiometer wire has length 4 m and resistance 6Ω. The resistance that must be connected in series with the wire and a battery of emf 4 V so as to get a potential gradient 5 mV per cm on the wire is

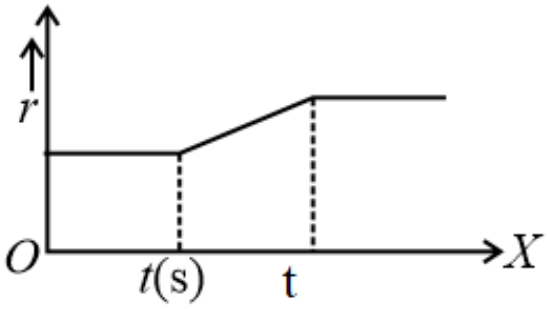
(A) 6Ω

(B) 12Ω

(C) 18Ω

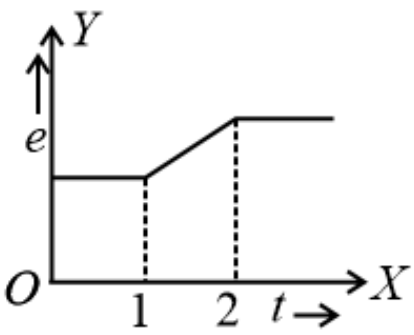
(D) 24Ω

Q.18



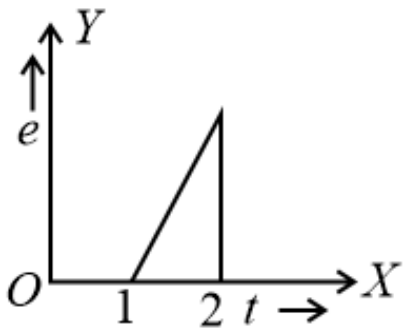
In a uniform magnetic field, a flexible wire is bent to form a circle with its plane perpendicular to the field. Above figure represents the change in the radius of the coil r with time t. Now, calculate the voltage induced in the coil:

(A)

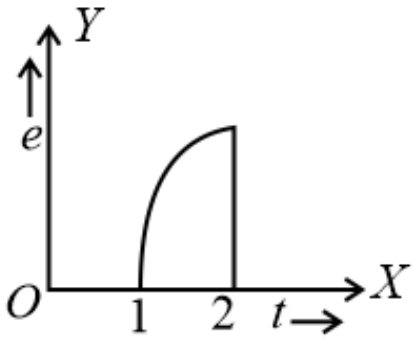


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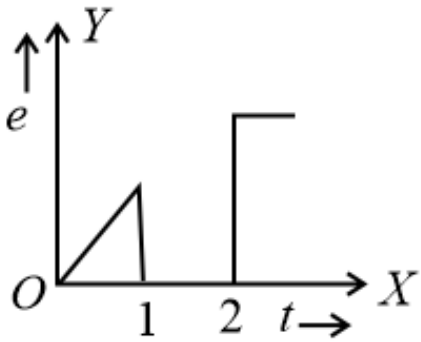
(B)



(C)



(D)



Q.19

The resistance of the series combination of two resistors is S. When they are joined in parallel, the equivalent resistance becomes P. If $S = nP$, then the minimum possible value of n is

(A) 3

(B) 4

(C) 5

(D) 6

Q.20

Two cylindrical wires A and B have the same resistance. The ratio of their specific resistances and diameters are 1: 2 each, then what is the ratio of the length of B to the length of A?

(A) 1

(B) 2

(C) 3

(D) 4

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Q.1 Answer:

1.4 ohm

Solution:

The network of resistances shown in Fig. 12.31 is not a balanced Wheatstone's bridge. In such a case, to find the resistance between A and B, we connect a battery of voltage V across A and B and find the current I drawn from the battery in terms of V and find the equivalent resistance from the relation $V = IR$. Refer to Fig.

The currents in various branches are shown in Fig. 12.32. Applying loops rule to loops $AFBGHA$, $ACDFA$ and $DEBFD$ we have

$$V - 1\,(I - I_1) - 2\,(I - I_1 + I_2) = 0 \quad \text{(i)}$$

$$2I_1 + I_2 - (I - I_1) = 0 \quad \text{(ii)}$$

and $(I_1 - I_2) + 2\,(I - I_1 + I_2) - I_2 = 0 \quad \text{(iii)}$

Simplifying these equations we get

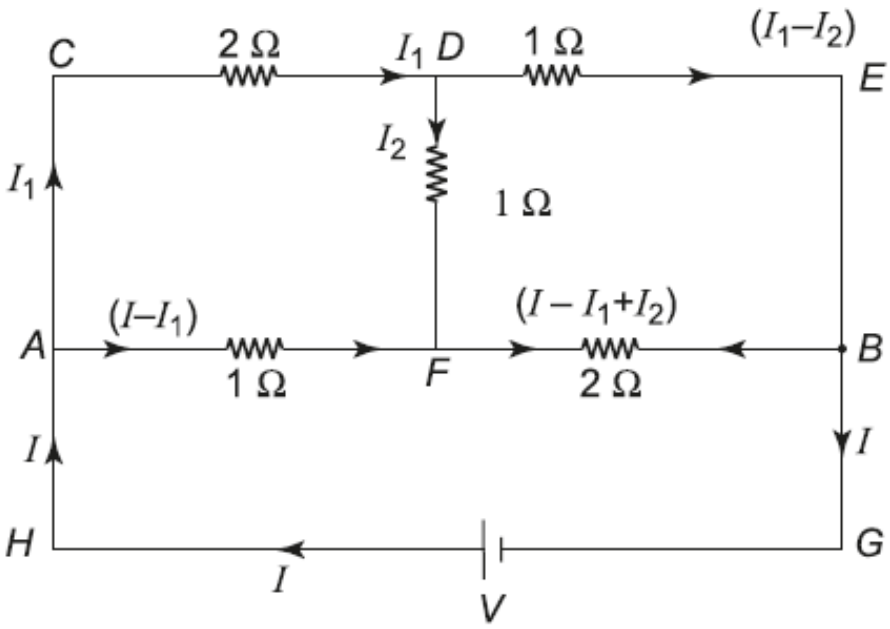
$$3I - 3I_1 + 2I_2 = V \quad \text{(iv)}$$

$$I - 3I_1 - I_2 = 0 \quad \text{(v)}$$

and $\quad 2I - 3I_1 + 4I_2 = 0 \quad \text{(vi)}$

Eliminating I_1 from (v) and (vi), we have $I_2 = -\frac{I}{5}$. Using this value of I_1 in (v) we get $I_1 = \frac{2}{5}I$. Using these values of I_1 and I_2 in (iv) we get

$$7I = 5\ \text{V} \Rightarrow \frac{V}{I} = \frac{7}{5} = 1.4 \Rightarrow R_{AB} = 1.4\Omega$$



Q.2 Answer:

$\frac{8}{3}$

Solution:

The resistance of a conductor of length l, crosssectional area A and made of a material of resistivity ρ is given by

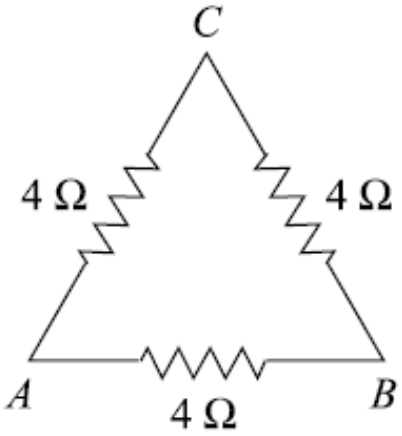
$$R = \frac{\rho l}{A} = \frac{\rho l^2}{Al} = \left(\frac{\rho}{V}\right) l^2$$

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where $V = Al$ is the volume of the conductor. Since ρ is a constant and volume V cannot change if the conductor is stretched, it follows that R is proportional to l^2. Thus if l is doubled, R becomes four times. Hence the new resistance is $3 \times 4 = 12\Omega$. So, each side of the equilateral triangle has a resistance of 4Ω as shown in Fig. 12.127. Therefore, the effective resistance between the ends of any side of the triangle (such as side AB) is equal to the resistance of a parallel combination of $R_1 = 4\Omega$ and $R_2 = 4 + 4 = 8\Omega$, which is given by

$R_e = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{4 \times 8}{4 + 12} = \frac{8}{3}\Omega$

Hence the correct choice is (b).



Q.3 Answer:

$5 : 4 : 3$

Solution:

Let the value of each resistance be r. The network can be redrawn as shown in the Fig.

(i) Net resistance R_1 between points A and B

The series combination of resistances $\frac{r}{3}$ and $\frac{r}{2}$,

which has an equivalent resistance $r_1 = \frac{r}{3} + \frac{r}{2} = \frac{5r}{6}$, is in parallel with resistance r. Hence

$R_1 = \frac{r \times r_1}{r + r_1} = \frac{r \times \frac{5r}{6}}{r + \frac{5r}{6}} = \frac{5r}{11}$

(ii) Net resistance R_2 between points B and C The series combination of resistances r and $\frac{r}{3}$,

which has an equivalent resistance $r_2 = r + \frac{r}{3} = \frac{4r}{3}$, is in parallel with resistance $\frac{r}{2}$. Hence

$R_2 = \frac{\frac{r}{2} \times r_2}{\frac{r}{2} + r_2} = \frac{\frac{r}{2} \times \frac{4r}{3}}{\frac{r}{2} + \frac{4r}{3}} = \frac{4r}{11}$

(iii) Net resistance R_3 between points A and C

The series combination of resistances r and $\frac{r}{2}$,

which has an equivalent resistance $r_3 = r + \frac{r}{2} = \frac{3r}{2}$,

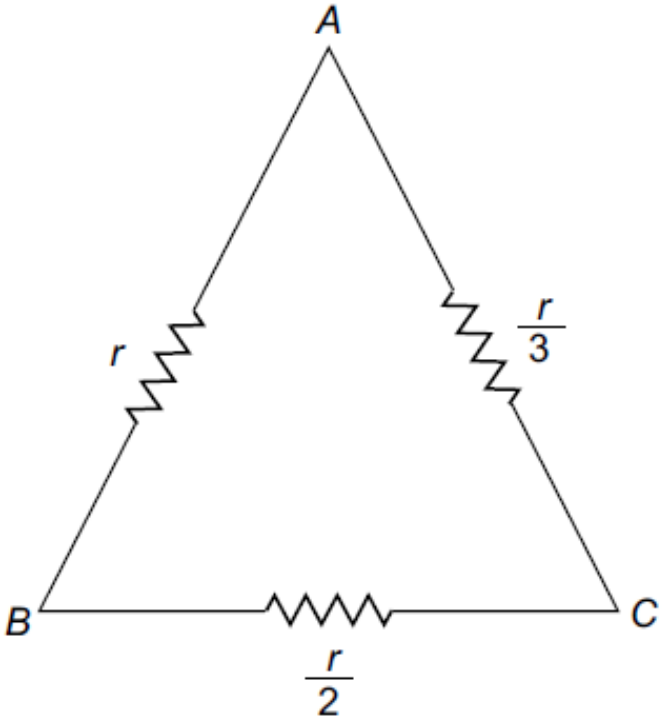
is in parallel with resistance $\frac{r}{3}$. Hence

$R_3 = \frac{\frac{r}{3} \times r_3}{\frac{r}{3} + r_3} = \frac{\frac{r}{3} \times \frac{3r}{2}}{\frac{r}{3} + \frac{3r}{2}} = \frac{3r}{11}$

Hence $R_1 : R_2 : R_3 = \frac{5r}{11} : \frac{4r}{11} : \frac{3r}{11} = 5 : 4 : 3$.

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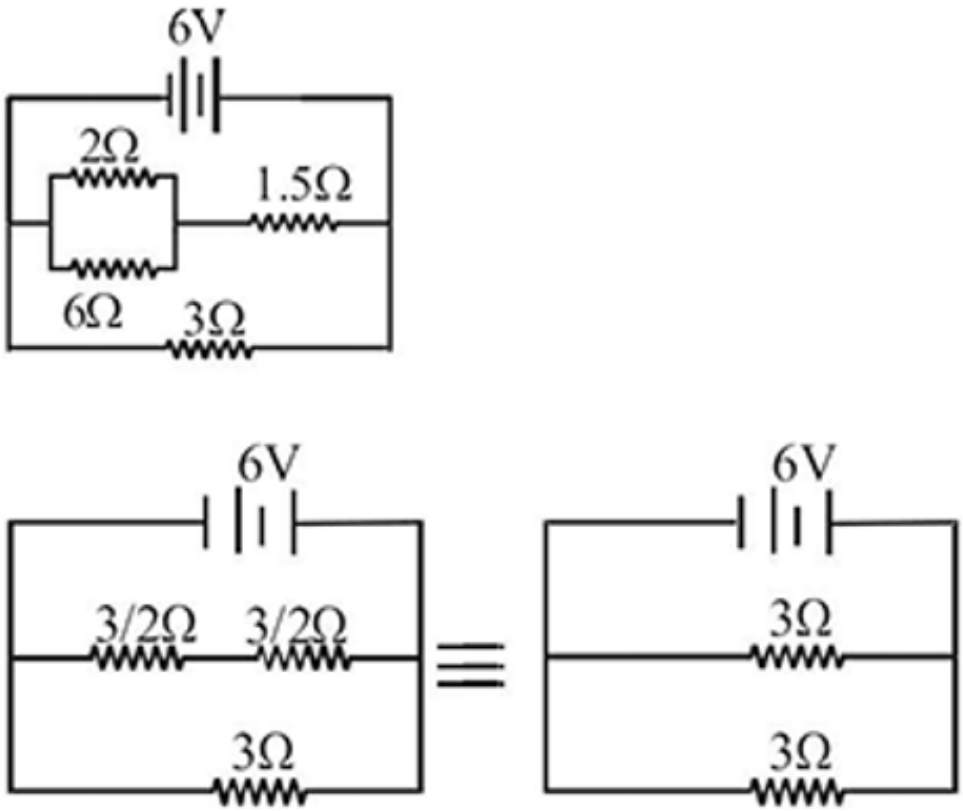
Thus the correct choice is (c).



Q.4 Answer:

4 A

Solution:



hence $R_{eq} = 3/2; \quad \therefore I = \frac{6}{3/2} = 4$ A

Q.5 Answer:

$R_A > R_B$

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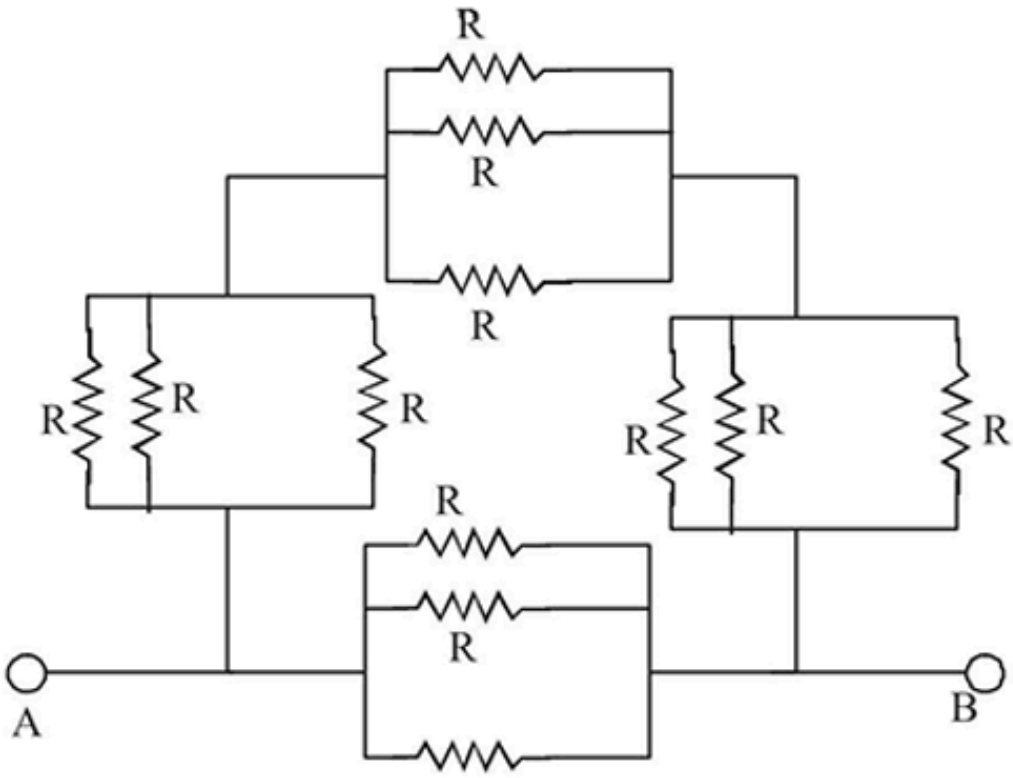
Solution:

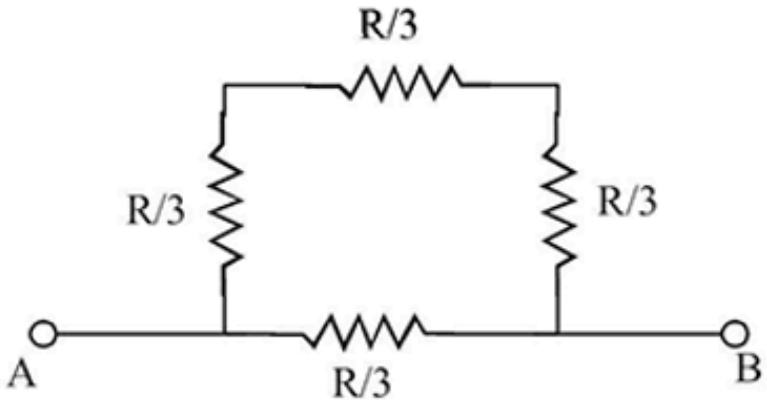
From the curve it is clear that slopes at points A, B, C, D have following order $A > B > C > D$. And also resistance at any point equals to slope of the $V - i$ curve. So order of resistance at three points will be $R_A > R_B > R_C > R_D$

Q.6 Answer:

4Ω

Solution:





R_{net} **between** $AB = \frac{\frac{3R}{3} \times \frac{R}{3}}{\frac{3R}{3} + \frac{R}{3}} = \frac{R^2}{4R} = 4\Omega$

Q.7 Answer:

n^2

Solution:

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In series, $R_s = nR$
In parallel, $\frac{1}{R_p} = \frac{1}{R} + \frac{1}{R} + \ldots n$ **term**
$\therefore R_s/R_p = n^2/1 = n^2$

Q.8 Answer:

$\frac{\mathrm{R}_0}{3}$

Solution:

By principle of symmetry and superposition, $2 \times \frac{\mathrm{I}}{6} \times \mathrm{R}_0 = \mathrm{IR}_{\text{eq.}} \Rightarrow \mathrm{R}_{\text{eq.}} = \frac{\mathrm{R}_0}{3}$
(Current $\frac{\mathrm{I}}{6}$ **in** AB **is due to division in current entering at A and current** $\frac{\mathrm{I}}{6}$ **is due to current returning from infinity of grid).**

Q.9 Answer:

the resistance will be halved and the specific resistance will remain unchanged

Solution:

$R = \frac{\rho \ell_1}{A_1}$, **now** $\ell_2 = 2\ell_1$
$A_2 = \pi \left(r_2\right)^2 = \pi \left(2r_1\right)^2 = 4\pi r_1^2 = 4A_1$
$\therefore \quad R_2 = \frac{\rho(2\ell_1)}{4A_1} = \frac{\rho \ell_1}{2A_1} = \frac{R}{2}$
$\therefore$ **Resistance is halved, but specific resistance remains the same.**

Q.10 Answer:

0.117

Solution:

Copper rod and iron rod are joined in series. $\therefore R = R_{\mathrm{Cu}} + R_{\mathrm{Fe}} = \left(\rho_1 + \rho_2\right) \frac{\ell}{A}$
$\left(\because R = \rho \frac{\ell}{A}\right)$
From ohm's law $V = RI = \left(1.7 \times 10^{-6} \times 10^{-2} + 10^{-5} \times 10^{-2}\right) \div$
0.01×10^{-4} **volt**
$= 0.117$ **volt** $\left(\because \mathrm{I} = 1 \mathrm{~A}\right)$

Q.11 Answer:

40Ω

Solution:

Length of each wire $= \ell$**; Area of thick wire** $\left(A_1\right) = 3$ **A; Area of thin wire** $\left(A_2\right) = A$ **and resistance of thick wire** $\left(R_1\right)^2 = 10\Omega$**. Resistance** $(R) = \rho \frac{\ell}{A} \propto \frac{1}{A}$ **(if** ℓ **is constant)** $\therefore \frac{R_1}{R_2} = \frac{A_2}{A_1} = \frac{A}{3A} = \frac{1}{3}$
or, $R_2 = 3R_1 = 3 \times 10 = 30\Omega$
The equivalent resistance of these two resistors in series $= R_1 + R_2 = 30 + 10 = 40\Omega$

Q.12 Answer:

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$\frac{8}{15}\mathrm{R}$

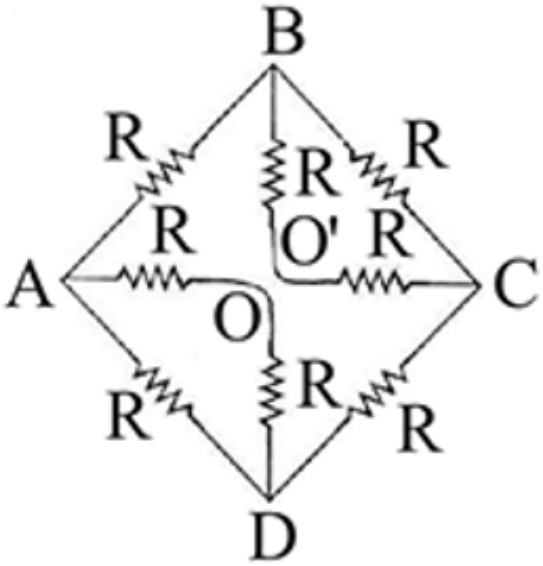
Solution:

The equivalent circuit is as shown in figure. The resistance of arm $\mathrm{AOD}(=\mathrm{R}+\mathrm{R})$ is in parallel to the resistance R of arm AD. Their effective resistance

$\mathrm{R}_1=\frac{2\mathrm{R}\times\mathrm{R}}{2\mathrm{R}\times\mathrm{R}}=\frac{2}{3}\mathrm{R}$

The resistance of arms AB, BC and CD is

$R_2=R+\frac{2}{3}R+R=\frac{8}{3}R$



The resistance R_1 and R_2 are in parallel. The effective resistance between A and D is $\mathrm{R}_3=\frac{\mathrm{R}_1\times\mathrm{R}_2}{\mathrm{R}_1+\mathrm{R}_2}=\frac{\frac{2}{3}\mathrm{R}\times\frac{8}{3}\mathrm{R}}{\frac{2}{3}\mathrm{R}+\frac{8}{3}\mathrm{R}}=\frac{8}{15}\mathrm{R}$

Q.13 Answer:

$0.2\ \mathrm{A}$

Solution:

Given : $V=7\nabla$ **and** $r=5\Omega$

$\mathrm{R}_{\mathrm{eq}}=\frac{40\times120}{40+120}\Omega$

$\mathrm{I}=\frac{\mathrm{V}}{\mathrm{R}}=\frac{7}{5+\frac{40\times120}{40+120}}=0.2\ \mathrm{A}$

Q.14 Answer:

2.1

Solution:

From color codes for resistors.

$R=21\times10^{8}\Omega\pm5\%=2.1\times10^{9}\Omega\pm5\%$

$\therefore\quad R=2.1G\Omega\pm5\%$

Q.15 Answer:

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$R_A R_v = G^2$ and $\frac{R_A}{R_v} = \frac{I_g}{I_0 - I_g}^2$

Solution:

$\left(I_0 - I_g\right) R_4 = I_g \mathrm{G}$ and $\mathrm{V}_0 = I_0 \mathrm{G} + \mathrm{R}_v$

$\mathrm{I}_0 = \frac{I_0 \mathrm{G} + \mathrm{R}_\mathrm{A}}{\mathrm{R}_\mathrm{A}} \quad \mathrm{V}_0 = I_0 \mathrm{G} + \mathrm{R}_\mathrm{V}$

$\mathrm{V}_0 = \mathrm{I}_0 \mathrm{G}\left(\text{ given }\right) \quad \Rightarrow \quad I_0 \mathrm{G} + \mathrm{R}_\mathrm{V} = \mathrm{I}_g \frac{\mathrm{G} + \mathrm{R}_\mathrm{A} \mathrm{G}}{\mathrm{R}_\mathrm{A}}$

$R_A G + R_V R_A = G^2 + R_A G$

$R_V R_A = G^2$

$R_A = \frac{I_g G}{I_0 - I_g}; R_V = \frac{I_0 - I_g G}{I_g}$

$\frac{R_A}{R_V} = \frac{I_g^2}{I_0 - I_g^2}$

Q.16 Answer:

9 Ω

Solution:

$\frac{R_1}{R_2} = \frac{20}{80} = \frac{1}{4}$, **It is clear that** R_1 **is smaller resistance**

$\therefore \quad R_2 = 4R_1$

When 15 Ω **is connected in series with** R_1**, then equivalent resistance is** $R_1 + 15$ **, So**

$\therefore \quad \frac{R_1 + 15}{R_2} = \frac{40}{60} = \frac{2}{3}$

$\therefore \quad \frac{R_1 + 15}{4R_1} = \frac{2}{3}$

$\therefore \quad R_1 = 9 \quad \Omega$

Q.17 Answer:

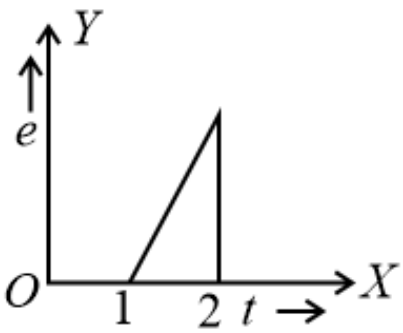
6Ω

Solution:

$L_{AB} = 4 \quad \mathrm{m} \qquad R_{AB} = 6\Omega \qquad E = 4 \quad \mathrm{V}$

$x = \frac{E}{R + R_s} \frac{R}{L} \Rightarrow \frac{5 \times 10^{-3}}{10^{-2}} = \frac{4}{R + 6} \frac{6}{4} \Rightarrow R = 6\Omega$

Q.18 Answer:



Solution:

The cross sectional area of a wire is given by $\pi \mathrm{r}^2$ **where** r **is the radius of the wire.**

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The emf induced in the wire is given by $\varepsilon = \frac{d\phi}{dt} = B\frac{dA}{dt} = B\frac{d}{dt}\pi r^2 = 2\pi Br\frac{dr}{dt}$

i.e., $\varepsilon \propto r\frac{dr}{dt}$

At $t = 0 - 1s$, $\frac{dr}{dt} = 0$. so $\varepsilon = 0$

The induced emf will be zero

At $t = 1 - 2$ s, $\frac{dr}{dt} = \text{constant}$. so $\varepsilon \propto r$

The graph will raise linearly.

At $t > 2s$, $\frac{dr}{dt} = 0$. so $\varepsilon = 0$

The curve will fall steeply to zero.

Q.19 Answer:

4

Solution:

In series combination, $S = R_1 + R_2$

In parallel combination, $P = \frac{R_1 R_2}{R_1 + R_2}$

$\because \quad S = nP$

$\therefore \quad R_1 + R_2 = n\frac{R_1 R_2}{R_1 + R_2}$

$\therefore \quad R_1 + R_2{}^2 = nR_1 R_2$

For minimum value, $R_1 = R_2 = R$

$R+R^2 = n\,R\times R \Rightarrow 4R^2 = nR^2 \Rightarrow n = 4$

Q.20 Answer:

2

Solution:

Resistance of a wire $R = \frac{\rho l}{\pi r^2} = \frac{\rho l \times 4}{\pi D^2}$

$\because \quad R_A = R_B$

$\therefore \quad \frac{4\rho_A L_A}{\pi D_A^2} = \frac{4\rho_B L_B}{\pi D_B^2}$

or $\frac{L_B}{L_A} = \frac{\rho_A}{\rho_B}\frac{D_B}{D_A}^2$

$= \frac{\rho_A}{2\rho_A}\frac{2D_A}{D_A}^2 = \frac{4}{2} = \frac{2}{1}$

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